Bangladesh University of Engineering and Technology
Department of Computer Science and Engineering

Course: CSE 206
Digital Logic Design Sessional

Experiment No. 1

Topic: Implementing circuit with basic gates.

Problem design and implementation:

Design and implement circuits for the following functions with basic gates. Do not simplify the circuits.

1. *F(X, Y, Z) = XY + Y'Z*
2. *F(X, Y, Z) = (X'YZ + XY'Z') (Z+X)*
3. *F(X,Y,Z)=X'+Y+Z'*

Lab sheet: For each of the problems/questions, your lab sheet should cover:

- Required IC chips
- Truth table
- Circuit diagram

Answer the following question:

1. A brief discussion on how you can construct XOR, NAND, and NOR gates using basic AND, OR, and NOT gates.